

Helicity Signals

Oct 28, 2009

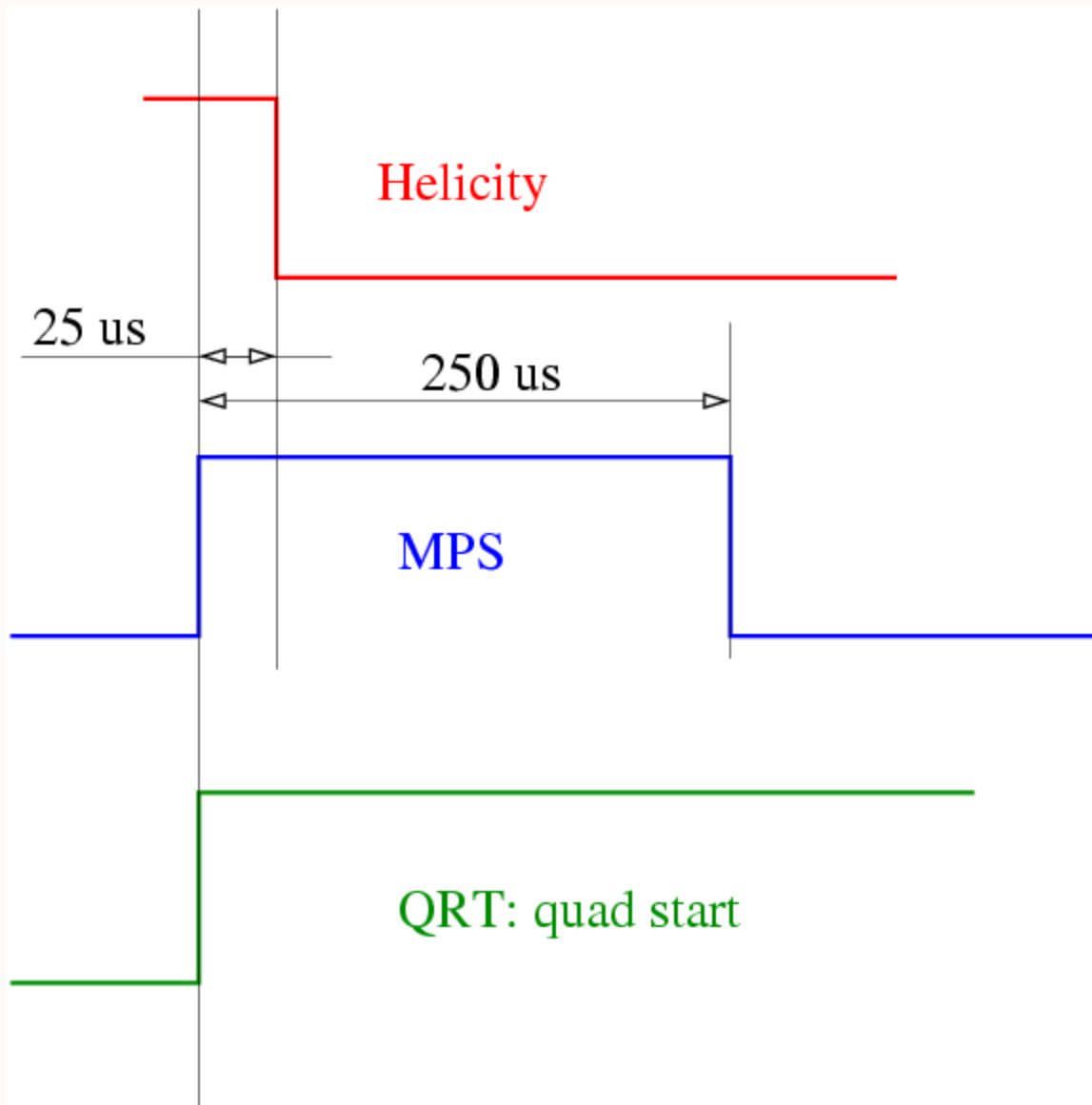
The time structure of the helicity signals has changed between June 2007 and April 2008. There are 3 signals:

1. Helicity signal in 33 ms long windows
2. MPS: a signal 250 μ s long, at the beginning of each helicity window
3. QRT: a 33 ms signal overlapping with the first-in-the-train helicity window. The train is normally a quadruplet of 33 ms windows

It turns out that the MPS is now comes 25 μ s earlier than the helicity flips. Before, they used to come synchronously. The QRT is synchronous with the MPS.

This happens to affect the Moller DAQ, since the helicity status is read out at the MPS front time. Before, it read out the NEXT window helicity with respect to the scaler information. Now, it is the same window helicity. The software can be adjusted to this change.

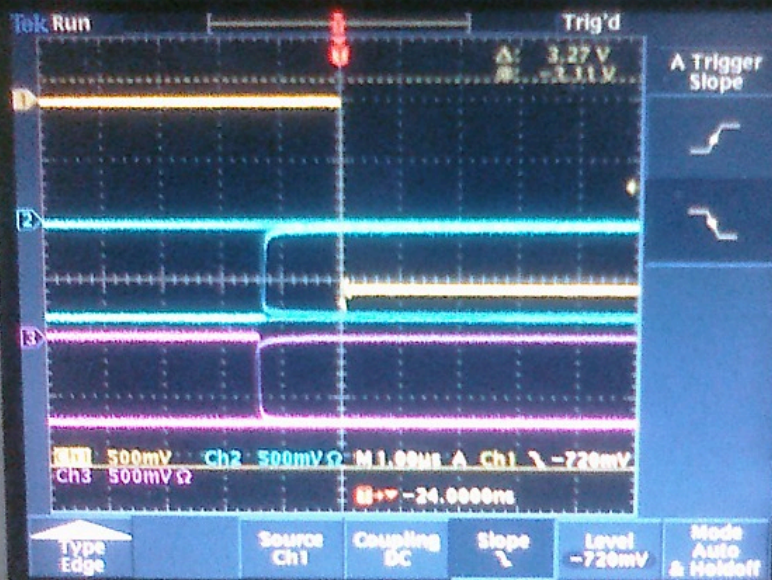
The present scheme might be more consistent than the old one.



On Oct 21 2009 Brad Sawatzky noticed that the signals changed again - the MPS is increased to 500us, the delay is reduced to 1.5us. Looking back at the data structure shows that it actually was changed during a break, sometime between May 27 and Aug 27 2009. The data changes - now the the scaler trigger latches the next helicity value. This did not affect the "scaler" analysis (asymwr.kumac), since by default the helicity is taken from the ADC triggers only.

Tektronix TDS 3054B

100MHz



Type Edge Source Ch1 Coupling DC Slope $\sim$ Level -720mV Mode Auto & Holdoff